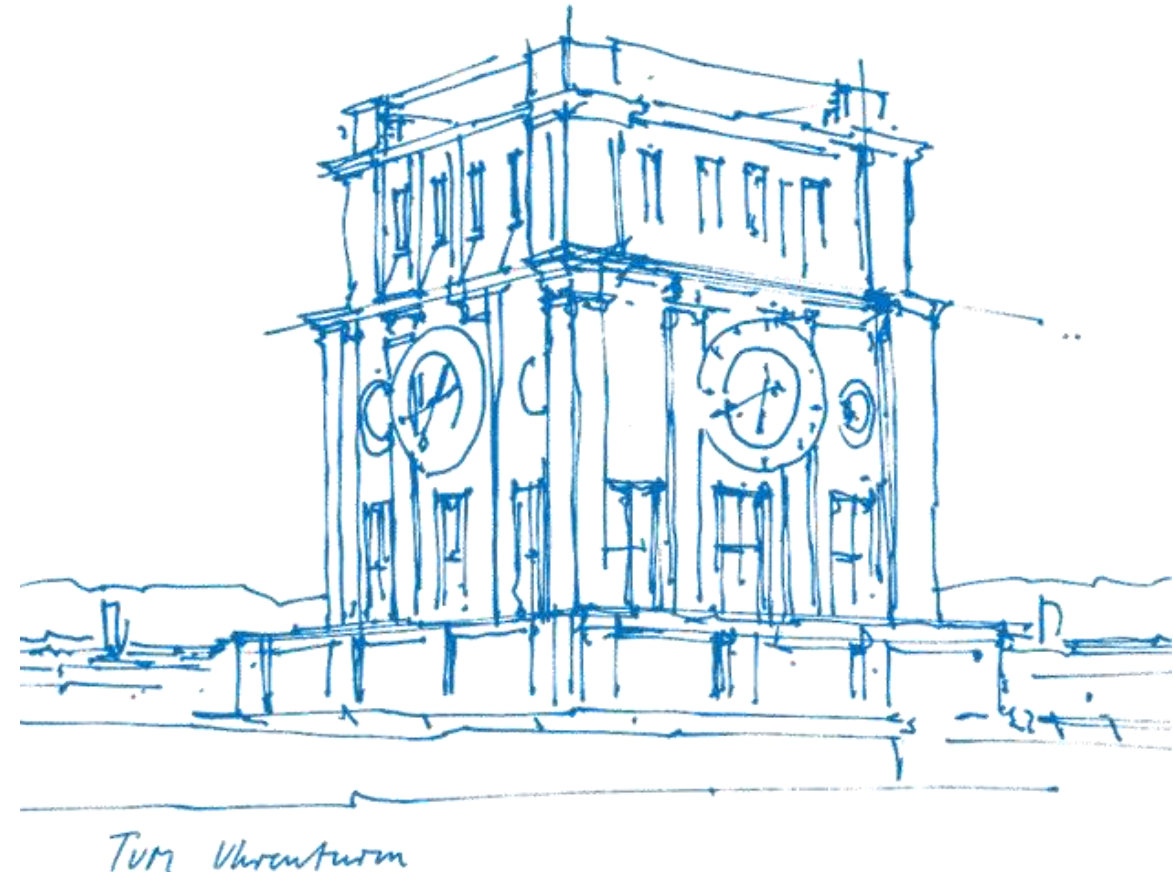


Tutorial 06

Expected Value, (Co)Variance, Independence and Uncorrelatedness

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8. June / 9. June 2026



About Tutorial

■ **Tutorial:** In-Person, 1.5h by two time periods per week - 12:15-13:45 Mon, 00.08.053 (A31)
- 14:15-15:45 Tue, 00.08.053 (B42)

■ The slides are not guaranteed to be accurate! (Written by myself)

■ **Materials of this tutorial:** Please scan the QR-Code 

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- **PART I.** Recap of the Lecture
- **PART II.** Exercises from the Exercise Sheet
- **PART III.** Additional Exercises (with solution)



I. Recap

- Expected Value, Variance and Covariance
 - Definition
 - Property
- Independence and Uncorrelatedness
 - Definition

<u>Distribution</u>	<u>Definition</u>	<u>Expectation</u>	<u>Variance</u>
<i>Bernoulli</i>	$f_X(x) = \begin{cases} p & x = 1 \\ 1 - p & x = 0 \end{cases}$	$\mathbb{E}[X] = p$	$\text{Var}[X] = p(1 - p)$
<i>Binomial</i>	$f_X(x) = \binom{n}{x} p^x (1 - p)^{n-x}$	$\mathbb{E}[X] = np$	$\text{Var}[X] = np(1 - p)$
<i>Geometric</i>	$f_X(i) = p(1 - p)^{i-1}$	$\mathbb{E}[X] = \frac{1}{p}$	$\text{Var}[X] = \frac{1 - p}{p^2}$
<i>Poisson</i>	$f_X(i) = \frac{e^{-\lambda} \lambda^i}{i!}$	$\mathbb{E}[X] = \lambda$	$\text{Var}[X] = \lambda$

II. Exercise Sheet

Exercise 1. Let $(\Omega, \mathcal{A}, P)$ be a probability space and $X, Y : \Omega \rightarrow \mathbb{R}$, $Z_1, Z_2, Z_3 : \Omega \rightarrow \mathbb{R}^n$ square integrable random variables/vectors and $a, b \in \mathbb{R}$. Verify the following computation rules for variance and covariance by using their definitions and the properties of the expectation we have mentioned in the lecture:

$$\text{var}[aX + bY] = a^2 \text{var}[X] + b^2 \text{var}[Y] + 2ab \text{cov}[X, Y]$$

and

$$\text{cov}[aZ_1 + bZ_2, Z_3] = a \text{cov}[Z_1, Z_3] + b \text{cov}[Z_2, Z_3] .$$

[Hint: You can use that for random vectors Z_1, Z_2 we have $\text{cov}[Z_1, Z_2] = \mathbb{E} [(Z_1 - \mathbb{E}[Z_1])(Z_2 - \mathbb{E}[Z_2])^\top]$, where $Z_1^\top$ is the transpose of Z_1 .

II. Exercise Sheet

Exercise 2. Calculate the mean and variance for

- (i) the discrete RV $X : \Omega \rightarrow \mathbb{N}$ with pmf $\rho(n) = \frac{\lambda^n}{n!} e^{-\lambda}$ for a fixed $\lambda > 0$.
- (ii) the binomially distributed RV $X : \Omega \rightarrow \{0, \dots, n\}$, meaning X models the number of successes observed in $n \in \mathbb{N}$ i.i.d trials where each trial is Bernoulli distributed with probability of success $p \in (0, 1)$.

[Hint: Consider writing $X = \sum_{i=1}^n Y_i$ for $Y_i \stackrel{\text{iid}}{\sim} \text{Ber}(p)$ and use Exercise 1 and Exercise 4.]

II. Exercise Sheet

Exercise 3. Let $(\Omega, \mathcal{A}, P)$ be a probability space and $X, Y : \Omega \rightarrow \mathbb{R}$ RVRVs such that the joint probability density function $p_{X,Y}$ (and thus also the marginal pdfs p_X, p_Y) exist. Further assume that expectations and (co-)variances of X, Y exist, i.e., X, Y are “square-integrable.” Show the following.

- (i) If X and Y are independent, they are uncorrelated.
- (ii) Give an example for X and Y being uncorrelated but not independent.
[Hint: Choose $Y = X^2$ for an appropriate choice of RV X . To show the non-independence, work with the cdfs.]

III. Additional Exercises

Exercise 4. Let $I_{A_1}, \dots, I_{A_n}$ be indicator variables for the pairwise distinct, independent events $A_1, \dots, A_n$, where $\Pr[A_i] = \frac{1}{n}$ for $i = 1, \dots, n$. What is the value of $\mathbb{E}[I_{A_1} \cdot \dots \cdot I_{A_n}]$?

—1

— $\frac{1}{n}$

— $\frac{1}{n!}$

✓ — $\frac{1}{n^n}$

$$E(I_{A_1} \cdot \dots \cdot I_{A_n}) = E(I_{A_1}) \cdot E(I_{A_2}) \cdot \dots \cdot E(I_{A_n})$$

$$= [1 \times \Pr(A_1) + 0 \times (1 - \Pr(A_1))] \cdot \dots \cdot [1 \times \Pr(A_n) + 0 \times (1 - \Pr(A_n))]$$

$$= \frac{1}{n} \cdot \dots \cdot \frac{1}{n}$$

$$= \frac{1}{n^n}.$$

Materials

